

Characterising Motion-Dependent Positioning Error in a Differential-Drive Robot for Improved Control and Motion-Strategy Selection

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Abstract—Accurate waypoint navigation in differential-drive robots is limited by odometric drift and motion-dependent heading bias, both of which accumulate across sequential stop-start cycles. This study characterises how motion primitive selection and aggressiveness setting influence final pose error in a Pololu 3Pi+ robot, and uses those findings to design an improved controller. Baseline experiments across straight-line, turn-in-place, and combined waypoint-seeking trials showed that lateral error grows quadratically with the number of intermediate stops ($R^2 > 0.99$), that arc motion consistently outperformed turn-then-drive at long range (4.7 mm vs 15.7 mm at $45^\circ/500$ mm), and that aggressive demand amplifies stopping error rather than in-motion error. An improved controller was designed that selects between arc and turn-then-drive based on target geometry, applies a trapezoidal deceleration ramp at the aggressive demand setting, and adds cross-track error correction to the drive phase. Evaluated on a four-waypoint circuit, the improved controller completed the route 10.2% faster than the turn-then-drive baseline and 4.9% faster than arc, while maintaining statistically equivalent position accuracy ($p > 0.79$), demonstrating that explicitly controlled stopping can improve speed without sacrificing accuracy.

I. INTRODUCTION

Autonomous mobile robots (AMRs) are becoming an increasingly important part of modern intralogistics systems. Robots are expected to move efficiently between pickup, drop-off, storage, and charging locations while operating in environments that may also contain other robots, obstacles, and changing task demands [1]. In such settings, successful navigation depends not only on planning feasible routes, but also on executing motion with sufficiently low final position and heading error [2]. This is especially important near docking or interaction regions, where inaccurate final alignment can lead to charging failures, operational delays, or material-handling errors [3].

This report focuses on that lower-level motion-execution problem using a Pololu 3Pi+ differential-drive robot. A differential-drive platform controls heading and speed by varying the relative velocities of two independently driven wheels, making it mechanically simple but sensitive to odometric drift and wheel slip - both of which accumulate over time and degrade target-reaching accuracy [4]. Although the experiments do not model the full sensing, coordination, and decision-making stack of industrial AMRs, they isolate a foundational requirement that remains relevant within any AMR deployment: reaching target waypoints accurately and repeatably.

A key question is whether different motion primitives produce different error characteristics. Paths with sharper turns can force a robot to slow down, whereas smoother arc trajectories may be preferable when motion feasibility and speed are considered together [5]. The aim of this study is therefore to characterise how straight-line motion, turning motion, and combined waypoint-seeking motion each contribute to final pose error, and then use that evidence to design a lower-error, motion-aware controller.

A. Hypothesis Statement

The Pololu 3Pi+ is a differential-drive robot whose heading changes are produced entirely by wheel-speed asymmetry, with no independent steering mechanism. Because turning accuracy is therefore directly limited by encoder resolution and wheel slip, three hypotheses have guided this investigation:

H1: *Different motion primitives produce different error characteristics, with turning-related motion contributing more strongly to final pose error than straight-line driving.*

H2: *More aggressive motion, particularly faster turning, increases both error magnitude and trial-to-trial variability.*

H3: *A motion-aware controller that selects between turn-then-drive and arc motion according to target geometry produces a measurable reduction in final pose error relative to a fixed baseline strategy.*

These hypotheses progress from a general characterisation of motion error (H1, H2) to a specific claim about controller design (H3), and are each reflected in the experimental methodology described in Section III.

The rest of this report proceeds as follows: Section II describes the hardware and software system used. Section III details the experimental methodology and metrics. Section IV presents the baseline error characterisation. Section V presents the improved controller results on a custom map, and Section VI discusses the conclusions drawn from the findings, addressing these hypotheses.

II. IMPLEMENTATION

A. Robot Platform

The experiments were conducted using a Pololu 3Pi+ differential-drive robot. Throughout the study, robot pose was represented as (x, y, θ) , where x and y denote planar position and θ denotes heading. The pose origin $(0, 0, 0)$ was defined at motion initialisation, so all reported positions and orientations are relative to the robot's starting pose at the beginning of each trial.

B. Odometry

Robot pose was estimated using onboard wheel-encoder odometry, following the differential-drive update scheme described in the course labsheets. At each update step, the change in encoder counts for the left and right wheels were converted into incremental wheel displacement. The mean of the two wheel increments was used to estimate forward displacement, while their difference was used to estimate the change in heading. These local increments were projected into the global frame using the midpoint heading over the update interval to update x , y , and θ . Wheel speeds were estimated from encoder count differences over a fixed sample period and low-pass filtered before use by the controller.

This configuration is self-contained and low-cost, requiring no external sensing. However, encoder-based odometry is vulnerable to wheel asymmetry and slip, particularly during turning and more aggressive manoeuvres [4]. These characteristics make it a suitable baseline for isolating the dominant contributors to final pose error.

C. Baseline Controller

The baseline controller used a nested architecture, shown in Figure 1. An outer pose-based loop generated wheel-speed demands, and inner PI loops regulated the measured left and right wheel speeds independently. Gains were tuned empirically and held fixed across all experiments, so that observed differences in error and completion time reflect motion primitive and aggressiveness setting rather than controller retuning.

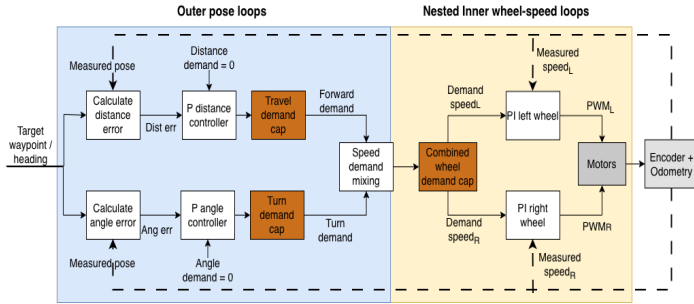


Fig. 1. Nested controller architecture for general motion. Encoder odometry feeds outer distance and angle loops; filtered wheel-speed estimates drive inner PI controllers. Left/right wheel-speed commands are derived via demand capping, mixing, and post-mix scaling.

Motion aggressiveness was varied by capping the forward and turn demands produced by the outer-loop controllers. Motion aggressiveness was varied through output capping rather than PID retuning, so that the closed-loop controller structure and gains remained fixed and any performance differences could be attributed primarily to the commanded motion intensity rather than to altered controller dynamics. For arc motion, the mixed left and right wheel-speed demands were uniformly scaled where necessary so that the larger of the two matched the selected aggressiveness limit, preserving their relative balance while bounding peak wheel speed. The outer angle controller derived a turning demand from heading error, defined as the wrapped angular difference between the robot's current heading and the bearing to the target waypoint.

D. Motion Primitives

Four motion primitives were implemented using the architecture above. *Arc-to-waypoint* applied forward and turning demands simultaneously to produce a curved trajectory towards the target. *Drive-straight* used only the distance branch, applying equal demands to both wheels. *Turn-on-spot* used only the angle branch, applying equal and opposite demands. *Turn-then-drive* applied these two reduced modes sequentially: first rotating to face the waypoint, then driving forwards.

Motion completion was detected from filtered wheel-speed feedback. A trial was considered complete when both filtered wheel speeds fell below 0.01 enc/ms, corresponding to a practically stationary robot without requiring an exact zero reading from the encoder measurements.

At completion, the controller logged the motion type, start pose, target pose, final pose, completion time, and odometric path length for later analysis.

III. EXPERIMENT METHODOLOGY

The 3Pi+ robot was tasked with reaching a series of target waypoints using different motion primitives and aggressiveness settings. The goal was to measure how motion type and aggressiveness influence final pose error, in order to evaluate the hypotheses stated in Section I.

A. Overview of Method

Three experiment groups were conducted: straight-line trials, turn-in-place trials, and combined-motion trials. Each group isolated a different contributor to final pose error. Two aggressiveness settings were used throughout: a *light* setting, in which forward and turn demands saturated at lower values, and an *aggressive* setting allowing faster motion. Each condition was repeated five times. Upon reaching each waypoint, the robot paused so that its final position and heading could be marked manually on paper, while serial logging simultaneously recorded the final odometric pose, completion time, and path length.

B. Discussion of Variables

- **Independent variables:** Motion primitive (drive-straight, turn-on-spot, turn-then-drive, arc-to-waypoint), aggressiveness setting (light or aggressive), and target geometry (distance and heading angle to waypoint).
- **Dependent variables:** Final position error (mm), final heading error (degrees), and motion completion time (ms), all logged at the end of each trial.
- **Controlled variables:** Controller gains, odometry update rate, motion-completion threshold, starting pose, floor surface, and battery state were held constant across all trials to prevent confounding sources of variation.

C. Experiment Conditions

- 1) **Straight-line trials.** The robot travelled a total of 1 m partitioned as five 200 mm segments, two 500 mm segments, or one 1000 mm segment. Each partition was repeated five times under both aggressiveness settings. Comparing equal total travel distance with different numbers of intermediate stops revealed whether forward position error, lateral drift, and completion time depend on segment length or accumulate over repeated stop-start cycles, addressing H2.
- 2) **Turn-in-place trials.** The robot achieved a total heading change of 180° using four sequential 45° left turns, two sequential 90° left turns, or one 180° left turn. Each sequence was repeated five times under both aggressiveness settings. A single 45° right turn was also tested to compare left- and right-turn behaviour. These trials measured heading bias, overshoot, and trial-to-trial variability, addressing H1 and H2.
- 3) **Combined-motion trials.** Four target cases compared *turn-then-drive* against *arc-to-waypoint*: targets at 250 mm and 500 mm, each paired with heading demands of 45° and 120°, giving short/small-angle, short/large-angle, long/small-angle, and long/large-angle cases. Each was tested under both aggressiveness settings for both primitives. The primary metrics were

final position error and completion time, as the task objective was accurate arrival at the waypoint rather than full pose regulation. These trials address all three hypotheses: whether motion type produces different error characteristics (H1), whether aggressiveness amplifies those differences (H2), and whether the preferable primitive depends on target geometry (H3).

D. Discussion of Metrics

Final position error was computed as the Euclidean distance between the target waypoint (x_t, y_t) and the manually marked final position (x_f, y_f) :

$$\epsilon_{pos} = \sqrt{(x_f - x_t)^2 + (y_f - y_t)^2} \quad (1)$$

Final heading error was the absolute wrapped angular difference between the target heading and the final logged odometric heading. Completion time was recorded in milliseconds from motion start to the moment both filtered wheel speeds fell below the stop threshold. Across the five repeated trials for each condition, the mean and standard deviation of each metric were used to characterise both typical performance and trial-to-trial variability. Where distributions overlapped, this was noted as evidence of similar performance; where they were well separated, this was taken as evidence of a meaningful difference between conditions.

IV. BASELINE ERROR CHARACTERISATION

A. Straight-Line Motion

Figure 2 and Table I show that total position error increases with the number of intermediate stops, where μ and σ denote the mean and standard deviation across five trials respectively. Under light motion, μ reduced from 80.7 mm at five stops to 4.8 mm for a single continuous segment, demonstrating that segmented and continuous motion produce fundamentally different error behaviours. The σ values indicate considerable trial-to-trial variability, suggesting the errors are not purely systematic.

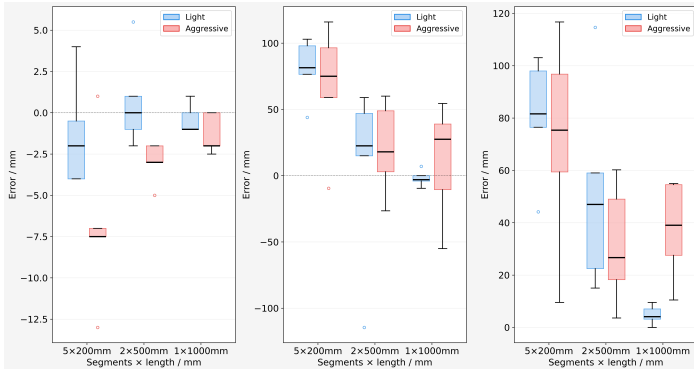


Fig. 2. Final position error across all three straight-line conditions under light and aggressive settings, shown as box plots ($n=5$ trials). Single points represent outliers.

TABLE I
MEAN (μ) AND STANDARD DEVIATION (σ) OF TOTAL POSITION ERROR FOR EACH STRAIGHT-LINE CONFIGURATION UNDER LIGHT AND AGGRESSIVE SETTINGS ($n=5$ TRIALS).

Configuration	Light		Aggressive	
	μ (mm)	σ (mm)	μ (mm)	σ (mm)
5×200 mm	80.7	23.2	71.6	40.9
2×500 mm	51.6	39.5	31.6	23.0
1×1000 mm	4.8	3.7	37.3	18.9

To characterise how error accumulates across stops, the 5×200 mm condition is examined in detail, as it provides five measurement points and therefore the clearest view of the accumulation trend. The 2×500 mm condition offers only two points, which is insufficient to characterise the growth pattern.

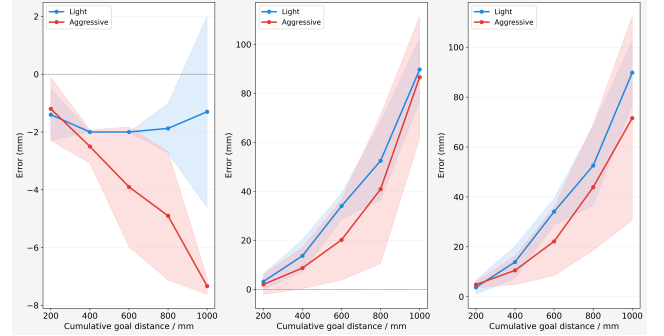


Fig. 3. Cumulative X, Y, and total position error across the five waypoints of the 5×200 mm condition ($n=5$ trials). Lines show μ and shaded regions indicate $\mu \pm \sigma$.

As shown in Figure 3, Y error grew from $\mu=3.2$ mm at 200 mm to $\mu=89.8$ mm by 1000 mm, following a quadratic growth pattern ($R^2 > 0.99$ under both settings), confirming that each stop-start event compounds the heading bias introduced by the previous one - a phenomenon of orientation error propagation - rather than introducing a fixed independent offset. X error showed no such accumulation under light motion, remaining approximately constant at -1.5 to -2.0 mm, whereas under aggressive motion it decreased linearly from -1.2 mm to -7.3 mm ($R^2 = 0.977$), indicating a systematic leftward drift consistent with wheel speed asymmetry at higher motor demands. The per-waypoint values are summarised in Table II.

TABLE II
PER-WAYPOINT μ OF X AND Y ERROR ACROSS THE 5×200 MM CONDITION ($n=5$ TRIALS).

Waypoint	Light		Aggressive	
	μ_Y (mm)	μ_X (mm)	μ_Y (mm)	μ_X (mm)
200 mm	3.2	-1.4	2.0	-1.2
400 mm	13.7	-2.0	8.7	-2.5
600 mm	34.0	-2.0	20.2	-3.9
800 mm	52.5	-1.9	40.9	-4.9
1000 mm	89.8	-1.3	86.6	-7.3

B. Turning Motion

Multi-step left-turn sequences produced a consistent undershoot bias, while larger single turns reversed the sign of the error. As shown in Figure 4, the 4×45° condition produced mean final errors of -2.6° under light and -1.4° under aggressive, whereas the 1×315° condition produced the largest errors overall, overshooting by $+3.4^\circ$ and $+3.2^\circ$ respectively.

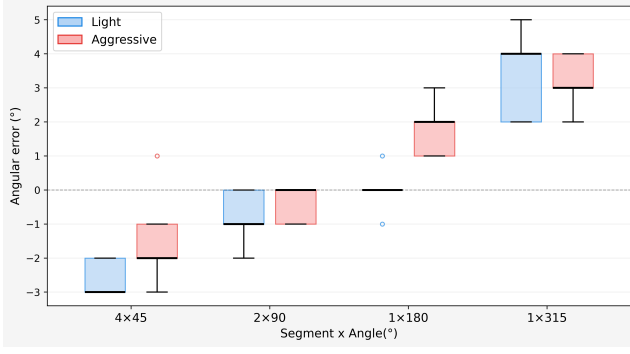


Fig. 4. Final angular error across all four turn conditions under light and aggressive settings, shown as box plots ($n=5$ trials). Single points represent outliers.

Figure 5 reveals how the bias develops across successive turns: the aggressive setting maintained a stable negative bias from the first turn (-0.4° at 45° , -1.6° at 90° and 135°), while the light setting began positive at $+1.7^\circ$ and drifted to -2.6° by 180° , suggesting the light controller oscillates around the heading setpoint before stopping. The $2 \times 90^\circ$ sequence showed a consistent overshoot-correction pattern - the first 90° turn was overshoot by $+2.6^\circ$ and $+2.2^\circ$ under light and aggressive respectively, with the second turn then undershooting - indicating the angle controller does not fully resolve heading error within a single turn sequence. Although angular errors were small in absolute terms ($1-4^\circ$), a 3° heading bias sustained over 500 mm of forward travel produces approximately 26 mm of lateral displacement, comparable to the straight-line errors observed above.

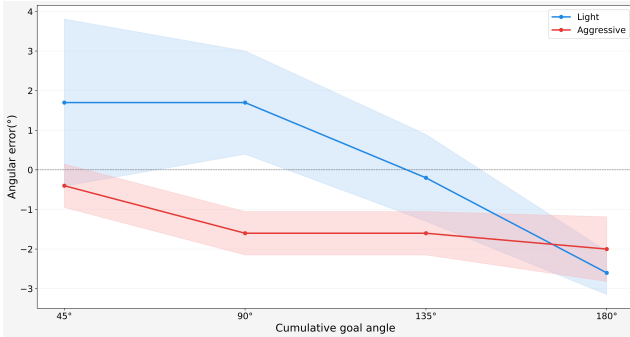


Fig. 5. Cumulative angular error across the $4 \times 45^\circ$ condition ($n=5$ trials). Lines show μ and shaded regions indicate $\mu \pm \sigma$.

C. Combined Waypoint-Seeking Motion

Arc motion produced substantially lower total errors than turn-then-drive across all conditions. For the $45^\circ/250$ mm case, arc motion achieved a mean total error of 1.1 mm under light compared to 5.8 mm for turn-then-drive. The gap was largest at $45^\circ/500$ mm, where turn-then-drive produced 15.7 mm under both settings while arc motion produced 5.6 mm under light and 3.8 mm under aggressive.

Figure 6 visualises this contrast spatially. Arc repeats cluster tightly near the target, while turn-then-drive repeats are displaced consistently in the positive-X/negative-Y direction. This systematic offset confirms the heading undershoot bias identified in the turning trials propagates directly into lateral displacement over the 500 mm drive phase.

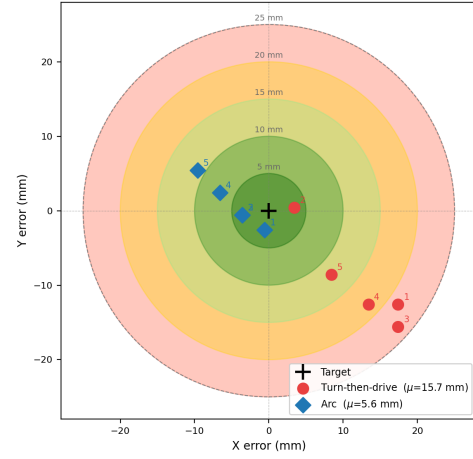


Fig. 6. Signed X-Y position error for the $45^\circ/500$ mm light condition ($n=5$ trials). Rings mark 5 mm increments. Arc (blue diamonds, $\mu = 5.6$ mm) clusters within 10 mm; turn-then-drive (red circles, $\mu = 15.7$ mm) sits beyond 15 mm with a consistent directional offset reflecting turn-phase heading undershoot.

Aggressiveness had little consistent effect on arc accuracy (Figure 7), while for turn-then-drive (Figure 8) the $120^\circ/250$ mm aggressive condition was an exception at just 2.9 mm, possibly due to turning overshoot partially compensating for the heading undershoot bias.

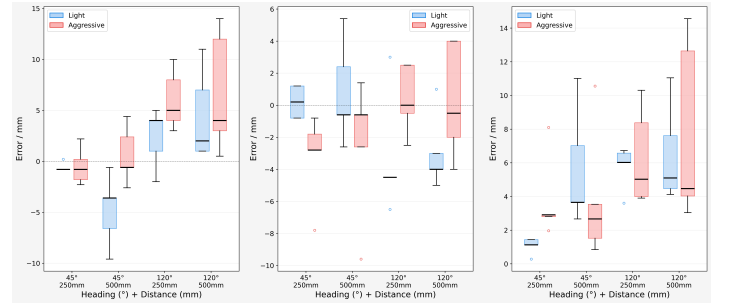


Fig. 7. Position error for arc-to-waypoint motion across all four target geometries under light and aggressive settings ($n=5$ trials). Single points represent outliers.

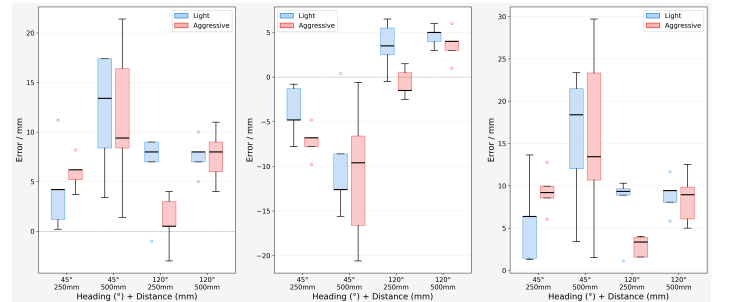


Fig. 8. Position error for turn-then-drive motion across all four target geometries under light and aggressive settings ($n=5$ trials). Single points represent outliers.

V. IMPROVED CONTROLLER RESULTS

A. Switching Logic

The primitive switching rule - turn-then-drive when distance < 200 mm, arc otherwise, with a heading-error override above 135° - was derived directly from the baseline combined-motion results. Arc outperformed turn-then-drive across every long-range condition: at $45^\circ/500$ mm,

arc produced $\mu = 4.7$ mm versus 15.7 mm for turn-then-drive; at $120^\circ/500$ mm, arc produced 5.6 mm versus 15.7 mm. The 200 mm distance boundary was chosen because arc accuracy degraded at 250 mm ($\mu = 1.1$ mm at $45^\circ/250$ mm, but 2.9 mm at $120^\circ/250$ mm), whereas turn-then-drive was competitive or better at short range in every condition. The 135° heading threshold reflects the observation that arc remained competitive at $120^\circ/500$ mm (5.6 mm) but becomes inefficient at larger angles where the curved convergence path adds substantial extra distance before the robot is aligned.

B. Trapezoidal Deceleration Ramp

Rather than reducing the demand ceiling to improve accuracy, the improved controller fixes demand at the aggressive setting (MAX_DEMAND= 0.75) and introduces a trapezoidal velocity profile to decouple speed from stopping accuracy. This is motivated by the baseline finding that the aggressive 1×1000 mm condition produced $\mu = 37.3$ mm error versus 4.8 mm under light - a difference attributed to abrupt stopping rather than to speed itself.

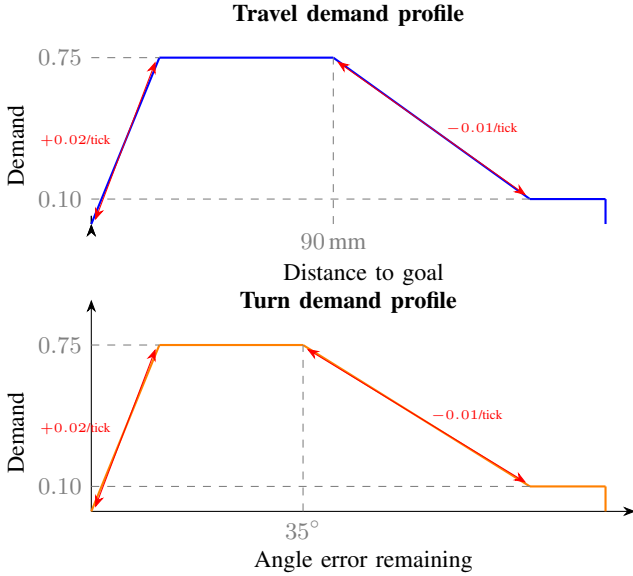


Fig. 9. Trapezoidal demand profiles for travel (top) and turn (bottom). Demand ramps up at +0.02 per PID tick (30 ms) and ramps down at -0.01 per tick within the deceleration window (90 mm for travel, 35° for turning), reaching the minimum demand of 0.10 before the settle threshold is checked. Both profiles share the same aggressive ceiling of 0.75.

The deceleration window values were chosen from baseline results: 90 mm for travel matches the longer stopping distance required at 0.75 demand, suppressing the 37.3 mm aggressive straight-line error; 35° for turning is wide enough to suppress the overshoot-then-correction pattern observed in the $2 \times 90^\circ$ condition ($+2.6^\circ$ first turn, corrected on second). The asymmetric ramp rates (+0.02 up, -0.01 down) produce a gentler deceleration than acceleration, reducing the heading kick at stop that baseline results identified as the primary driver of quadratic Y-error accumulation across stop-start cycles.

C. Position Error

The improved controller was evaluated on a four-waypoint circuit: $(0, 0) \rightarrow (212, 212) \rightarrow (212, 112) \rightarrow$

$(283, 41) \rightarrow (283, 341)$, repeated five times alongside baseline turn-then-drive and arc runs on the same circuit. The switching rule selected arc for WP0 and WP3 (≈ 300 mm legs) and turn-then-drive for WP1 and WP2 (≈ 100 mm legs), exercising all four geometry cases from the baseline experiments. Overall mean total errors were 10.18 mm for turn-then-drive, 9.23 mm for arc, and 9.67 mm for the improved controller. Paired t -tests confirm no statistically significant difference between the improved controller and either baseline (vs. turn-then-drive: $t = -0.269$, $p = 0.793$; vs. arc: $t = 0.194$, $p = 0.850$), indicating accuracy equivalent to both baselines.

TABLE III
PER-WAYPOINT MEAN (μ) AND STANDARD DEVIATION (σ) OF TOTAL POSITION ERROR ($n = 5$ TRIALS). PRIMITIVE SELECTED BY THE IMPROVED CONTROLLER SHOWN IN PARENTHESES. ALL VALUES ARE IN MM.

Waypoint	Turn-then-drive		Arc		Improved	
	μ	σ	μ	σ	μ	σ
WP0 (arc)	15.0	4.3	8.5	1.2	12.3	4.5
WP1 (turn-then-drive)	5.2	3.0	6.7	1.4	8.0	2.3
WP2 (turn-then-drive)	8.4	1.8	9.8	1.5	5.7	4.4
WP3 (arc)	12.1	5.7	11.9	3.5	12.7	11.5

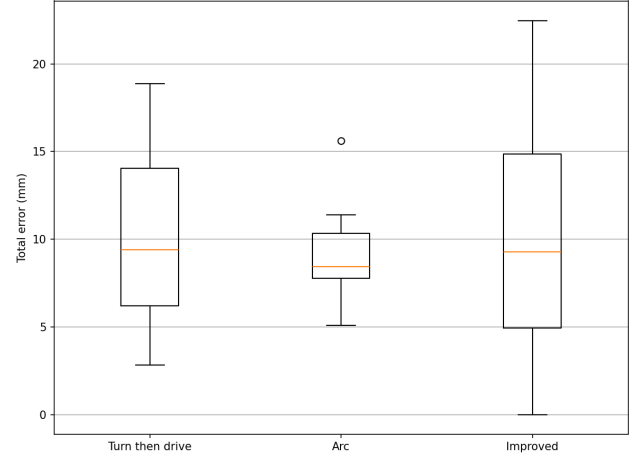


Fig. 10. Distribution of total position error pooled across all waypoints and repeats ($n = 5$ per method). The improved controller achieves a comparable median error to both baselines while operating at the aggressive demand setting.

Figure 10 shows the improved controller error distribution sits between the two baselines. The cross-track correction is most visible at WP2 (5.7 mm improved vs 8.4 mm baseline turn-then-drive), where the short diagonal leg previously produced consistent lateral drift that heading-to-goal correction alone could not suppress. The elevated WP3 variability ($\sigma = 11.5$ mm) is attributed to accumulated odometry error from the three preceding motions rather than a controller deficiency.

D. Speed and Consistency

Figure 11 jointly encodes speed, accuracy, and trial-to-trial consistency. Mean circuit times were 10,321 ms for turn-then-drive, 9,744 ms for arc, and 9,266 ms for the improved controller - 10.2% and 4.9% faster respectively - with a timing standard deviation of 43 ms, the lowest of all three methods. The speed gain over turn-then-drive comes from eliminating the sequential turn phase on long legs; the gain over arc comes from running at aggressive demand with the trapezoidal ramp preventing the stopping errors

observed in the aggressive baseline straight-line trials. The common mid-circuit drop across all three methods is mainly a waypoint-layout effect, since WP1 and WP2 are shorter legs and therefore accumulate less translational and heading error than WP0 and WP3.

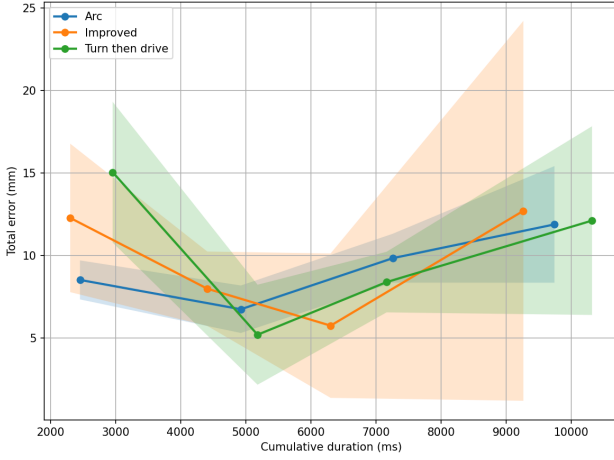


Fig. 11. Mean total error $\pm \sigma$ band at each waypoint against cumulative completion time ($n = 5$ trials). The improved controller reaches each waypoint sooner than both baselines with comparable or narrower error bands through WP2; the wider WP3 band reflects accumulated odometry drift.

E. Discussion

The improved controller achieves its primary objective - faster circuit completion with statistically equivalent accuracy - confirming all three baseline hypotheses. H1 is addressed by the switching logic selecting the better primitive per geometry, directly exploiting the performance gap between arc and turn-then-drive quantified in the baseline. H2 is resolved by the deceleration ramp decoupling speed from stopping accuracy, allowing aggressive demand without the error penalty observed in baseline trials. H3 is validated directly: turn-then-drive is selected and performs better at short range (6.9 mm mean vs 8.3 mm arc); arc is selected and performs better at long range (12.5 mm mean vs 13.6 mm turn-then-drive for the same legs).

VI. DISCUSSION AND CONCLUSION

This study characterised how motion primitive selection and aggressiveness setting influence final pose error in a differential-drive robot, and used those findings to design an improved motion-aware controller.

H1 - that different primitives produce different error characteristics - was confirmed. Arc motion consistently outperformed turn-then-drive at long range (4.7 mm vs 15.7 mm at $45^\circ/500$ mm), while turn-then-drive was competitive at short range. Straight-line trials isolated orientation error propagation as the primary mechanism: Y error grew quadratically across stop-start cycles ($R^2 > 0.99$), confirming that heading bias compounds rather than accumulates independently.

H2 - that aggressiveness amplifies error - was partially confirmed. Under the aggressive setting, the 1×1000 mm condition produced $\mu = 37.3$ mm versus 4.8 mm under light, and X drift grew linearly to -7.3 mm, consistent with wheel-speed asymmetry at higher demands. However, aggressiveness had little consistent effect on arc accuracy,

indicating that the error penalty is specific to abrupt stopping rather than to speed per se - a distinction directly exploited by the improved controller.

H3 - that a geometry-aware controller reduces error - was confirmed in terms of speed, though not in mean position error. The improved controller completed the circuit 10.2% faster than turn-then-drive and 4.9% faster than arc, with a timing standard deviation of 43 ms - the lowest of all three methods - while maintaining statistically equivalent accuracy ($p > 0.79$ against both baselines). The deceleration ramp successfully decoupled speed from stopping accuracy, and the cross-track correction reduced WP2 error from 8.4 mm to 5.7 mm by suppressing lateral drift that heading-to-goal correction alone could not address.

Limitations and Reproducibility

Several limitations affect the generalisability of these results. All trials were conducted on a single floor surface with a fixed battery state; the consistent leftward X drift (-7.3 mm at 1000 mm aggressive) suggests wheel asymmetry that may vary across robot units and charge levels, reducing reproducibility across hardware. Although the controller tuning is applicable to all differential-drive robots, the performance was not evaluated for other robots, and therefore, generalisation cannot be assumed. The odometry-only pose estimation accumulates drift across sequential waypoints, as reflected in the elevated WP3 variability ($\sigma = 11.5$ mm), and provides no mechanism for external correction.

Future Work

The most impactful extension would be integrating the onboard IMU available on the 3Pi+ to fuse gyroscope measurements with encoder odometry, directly reducing the heading bias that drives quadratic Y-error accumulation [6]. Adding an integral term to the outer angle controller ($K_i > 0$) would address the residual turn undershoot bias (-2.6° under light) without hardware changes. The switching threshold of 200 mm was derived empirically from a limited condition set; a more systematic calibration across a wider range of distances and angles would produce a more robust decision boundary. Finally, extending the circuit to more waypoints would stress-test whether the deceleration ramp prevents the error compounding observed in the 5×200 mm baseline at scale.

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